

A low-angle, upward-looking perspective of several tall skyscrapers. The image is heavily stylized with a red color overlay, particularly on the right side and top. The buildings are dark, and the sky is a solid red. The text is white, providing a high contrast against the red background.

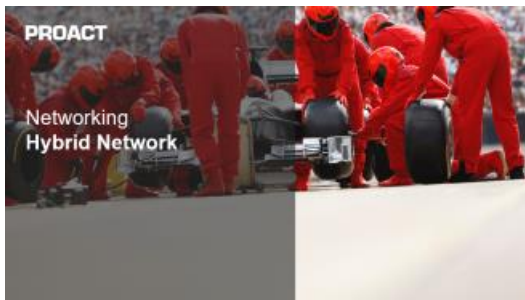
Workshop
NETWORKING

PROACT

Azure Network Session

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Agenda



About the Azure Virtual Networks

❑ Address spaces

This is the address space of the network which can be assigned to subnets created within the subnet

❑ Subnet

A segment within the virtual network address space which can be used by resources like virtual machines, application gateway, private link

❑ Regions

A virtual network is scoped to a single region. If you deploy a VM in EastUS2, the VNET should also be in EastUS2

❑ Subscription

A virtual network is scoped to a subscription, only resources within the subscription (and region) can use the virtual network.

See [Azure docs](#) for more information

Best practices

- ✓ Ensure non-overlapping address spaces. Make sure your VNET address space (CIDR block) does not overlap with your organization's other network ranges.
- ✓ Your subnets should not cover the entire address space of the VNET. Plan ahead and reserve some address space for the future.
- ✓ It is recommended you have fewer large VNETs rather than multiple small VNETs. This will prevent management overhead.
- ✓ Secure your VNETs by assigning Network Security Groups (NSGs) to the subnets beneath them.

Virtual Network Limitations

- ❑ Virtual Network Address spaces cannot be updated when subnets are deployed.
- ❑ Additional address spaces can be added to a virtual network but requires the removal of the peerings / virtual wan hub connections
- ❑ Subnets cannot be split/merged or stretched when resources are already connected to the subnet
- ❑ Maximum depth level Management Groups is 6
- ❑ Only 1 VPN Gateway per VNET
- ❑ Only 1 Expressroute per Virtual WAN

Other limitations apply and can be found [here](#)

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Subnets within LZ (to be discussed)

- ❑ Who **controls on the network segmentation of the subnets** within the landing zone.
 - ❑ Delegated to landingzone owners, responsibility of connectivity team limited
 - ❑ Connectivity team designs and implements the subnetting within the landingzone

- ❑ Do we want to prescribe a default set of subnets per landingzone or do we design on demand
 - ❑ Default subnet (private / public or other)
 - ❑ Leave room for specialized resources which require dedicated subnets (like SQL Managed Instance, Application Gateway, Azure Firewall, Bastion, Private link)

Delegation of control or managed by connectivity



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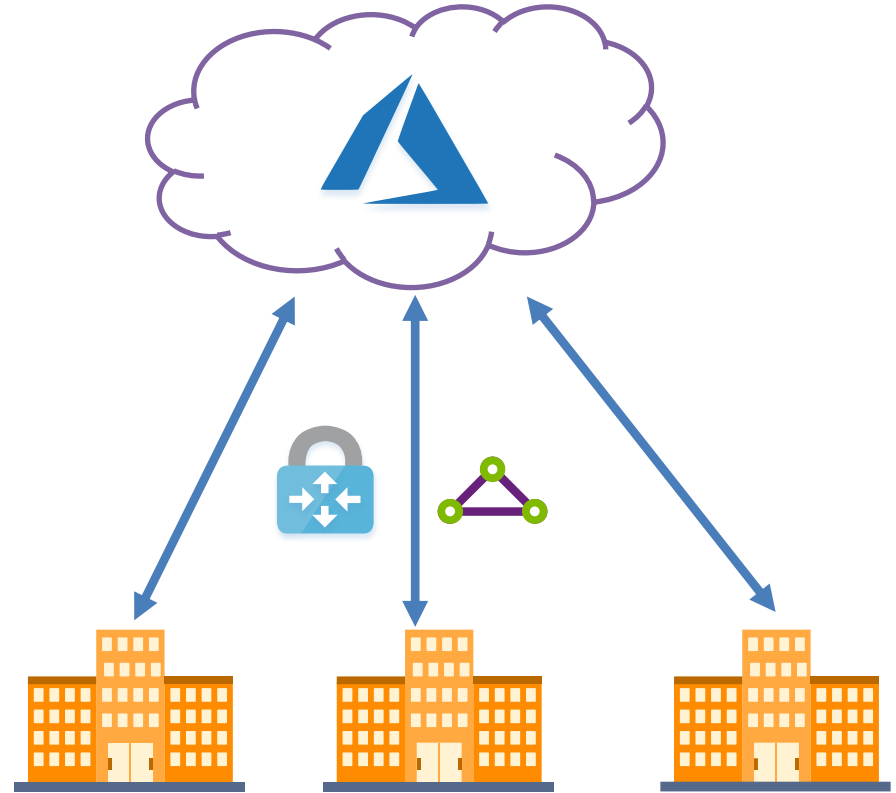
Networking
Hybrid Network

Hybrid networking

- Allows private communication between
 - » On-premise datacenter
 - » Branch offices
 - » Azure Virtual networks

Discussion

- Which branches / datacenters should communicate with Azure
 - » Private (IaaS)
 - » Public (PaaS)



Hybrid Connectivity

Connectivity option

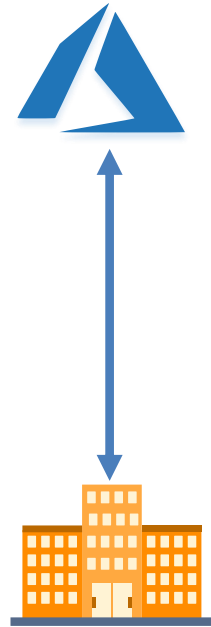
- Allow from On-premise to Azure only
- Allow Bidirectional between On-premise and Azure
- Provide WAN between On-premise locations and Azure (full mesh)

Discussion

Which services will rely on services on-premises?

Which services will rely on services in Azure?

How do these services communicate (internal network or internet)



Connectivity Options



S2S VPN

- Up to 1.5 Gbs
- Encrypted over the internet
- Recommended for most use-cases

Considerations

- Shared bandwidth (internet)
- Higher latency (~10-30ms)
- Easy/fast to setup
- Affordable



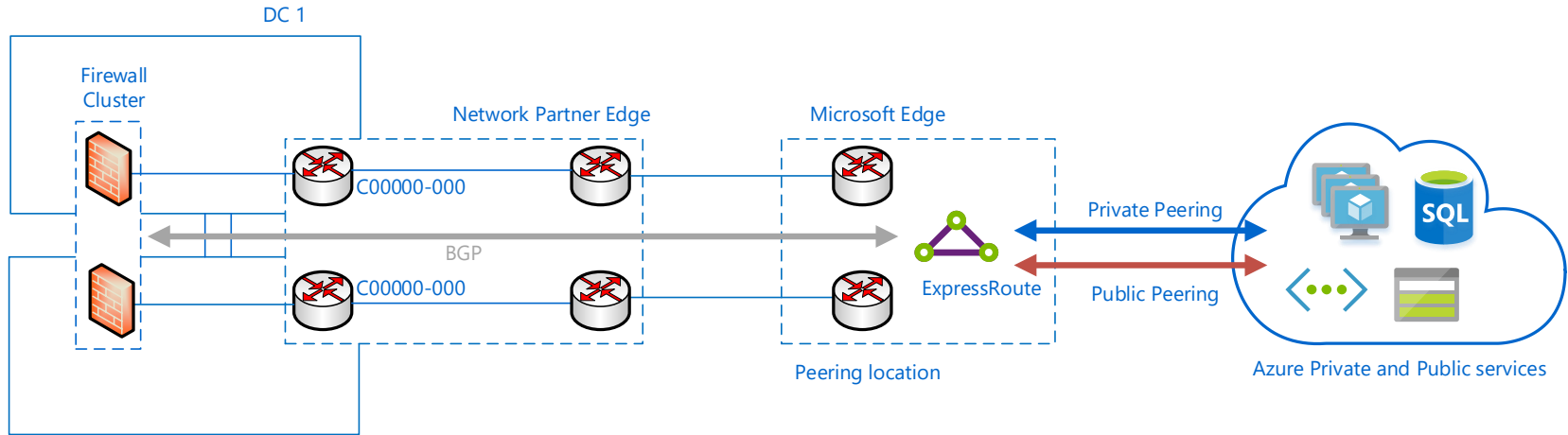
ExpressRoute

- Up to 100Gbs
- Over dedicated connection
- Recommended for low-latency long term connectivity

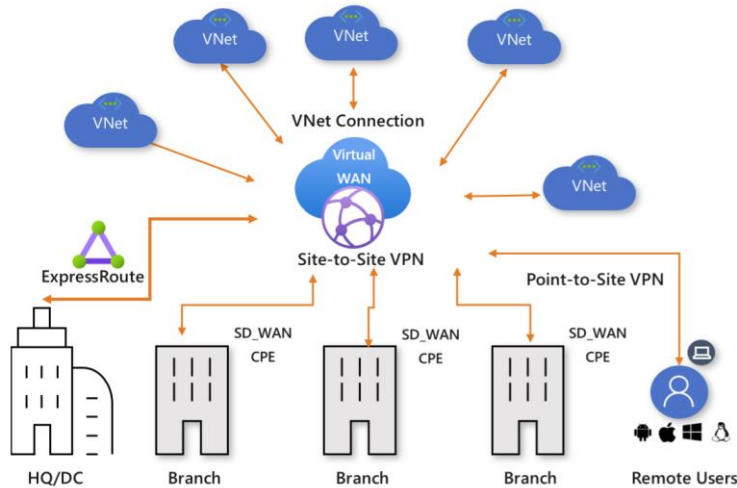
Considerations

- Dedicated bandwidth
- Lower latency (~3-10 ms)
- Complex/lengthy to setup
- Costly (provider + circuits)

ExpressRoute



Virtual WAN



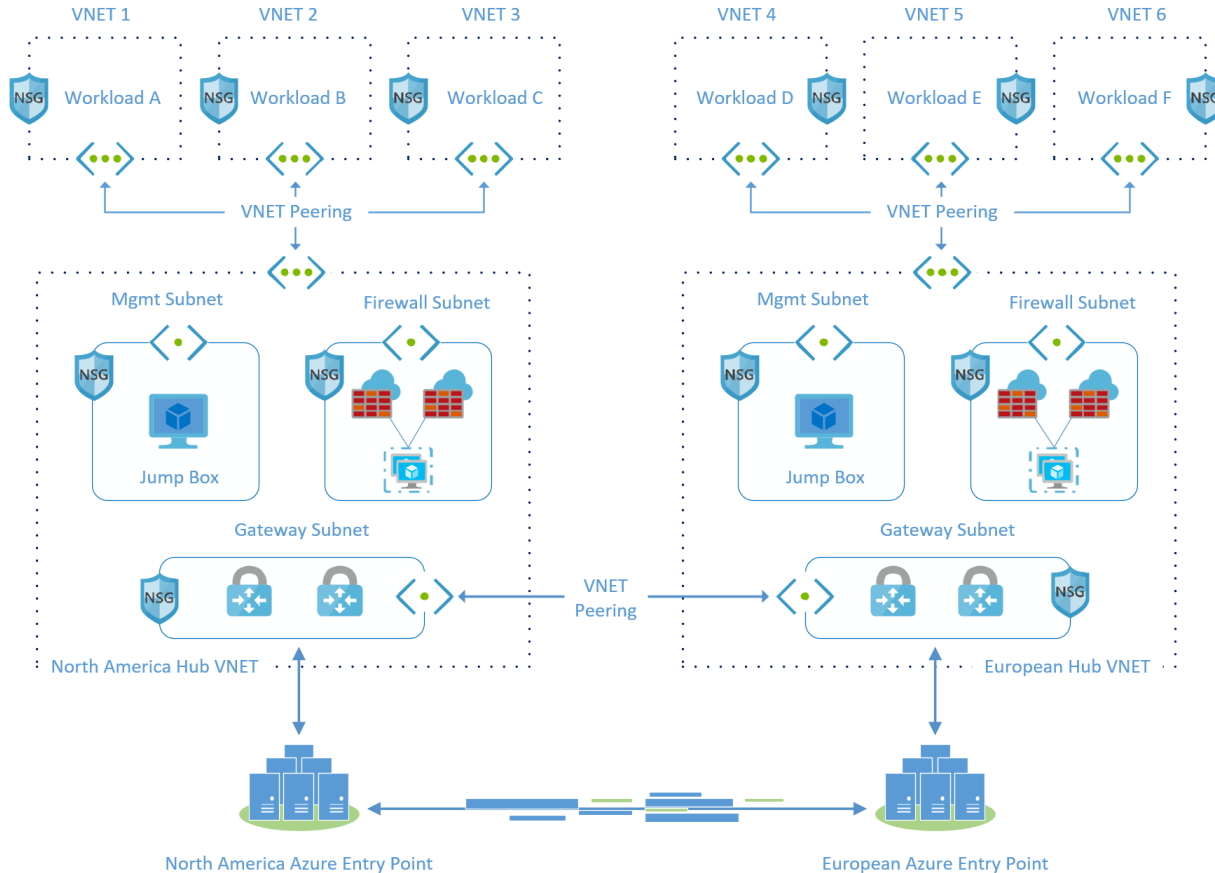
- Replacement of existing WAN
- Branches connected over VPN
- P2S available for workplace
- Firewall (optional)
- Auto configuration with connectivity partners

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Multiregion Hub and Spoke



Multi-region Hub and Spoke



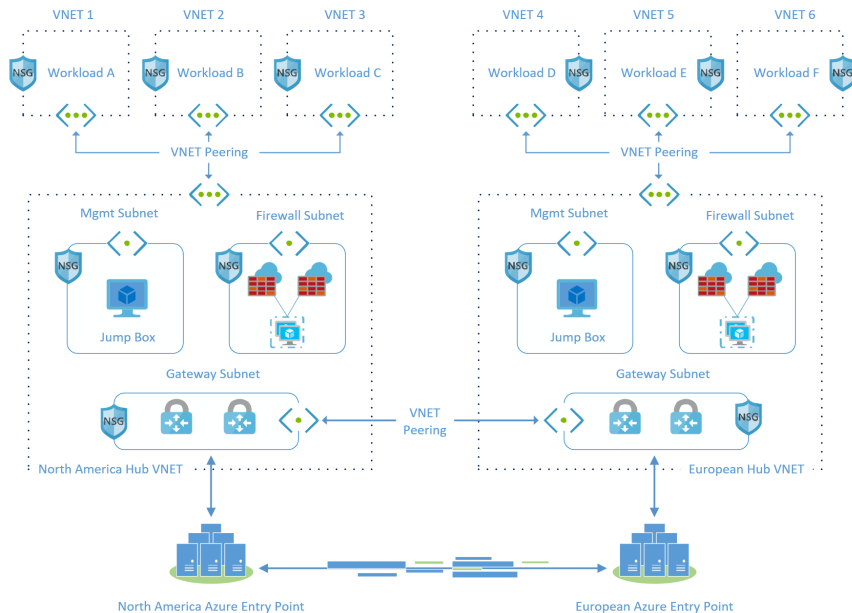
Global traffic

- Routing traffic between services in Azure
- Services which will be shared amongst regions

Discussion

Is traffic between regions required?

Should Workload A connect internally with Workload D?



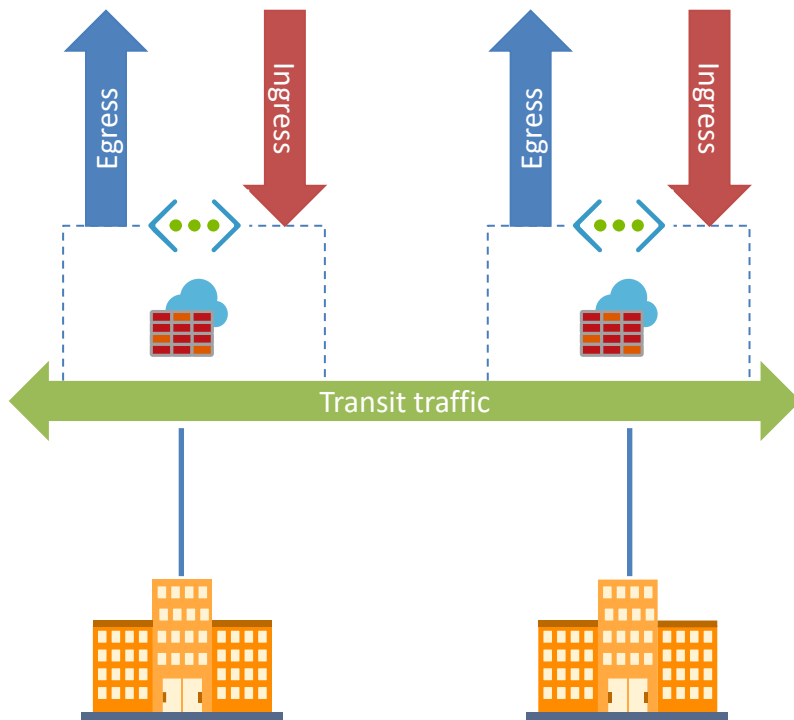
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Networking
Ingress / Egress /
Transit traffic



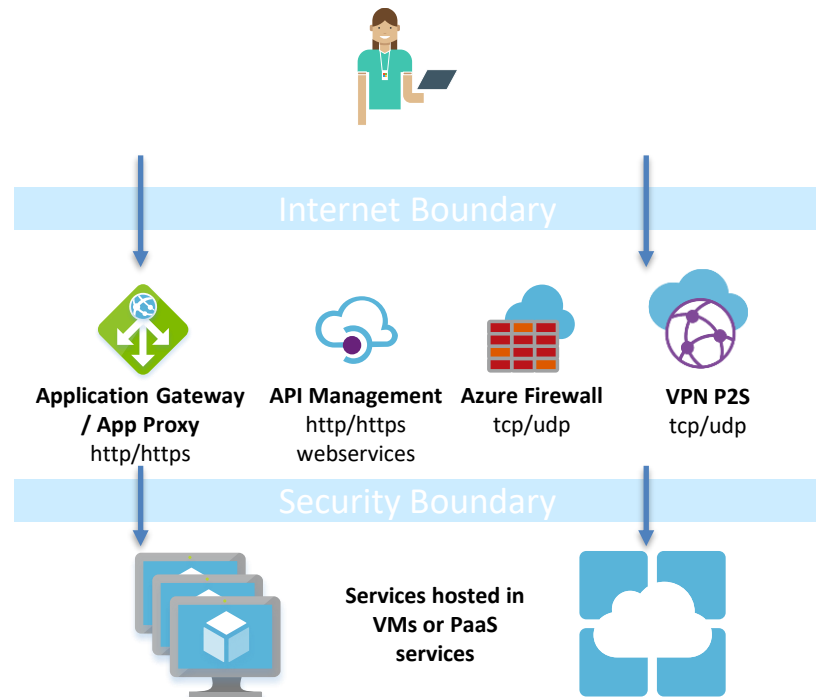
Traffic

- Ingress traffic
 - » Inbound internet traffic to Azure
- Egress traffic
 - » Outbound internet traffic from Azure
- Transit traffic
 - » Inter-region traffic



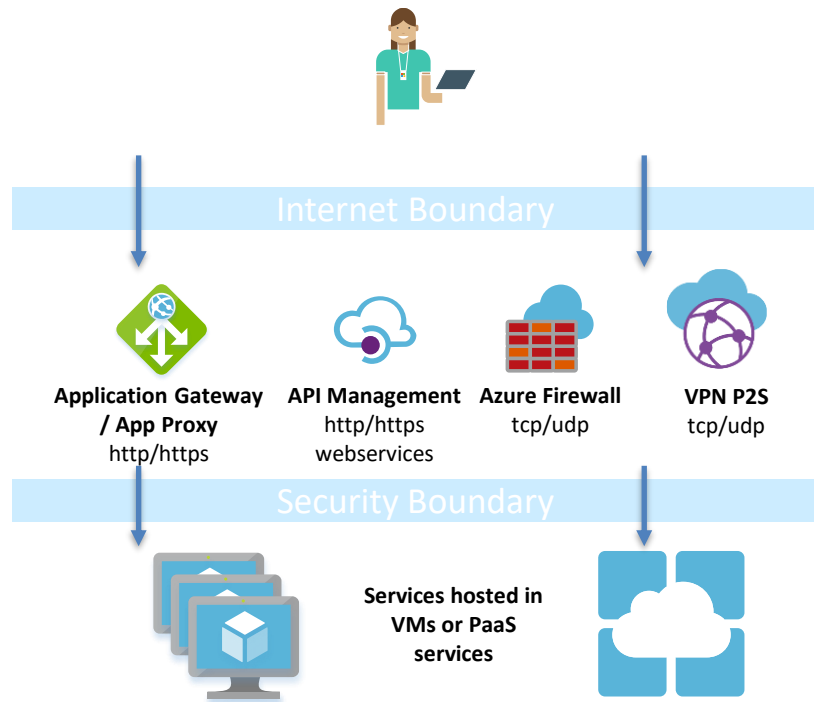
Ingress traffic

- IaaS virtual network
 - » Azure Firewall
 - » Application Gateway / WAF
 - » Azure AD App Proxy



Ingress traffic

- PaaS public service
 - » Application gateway for http/https
 - » API Management for webservices
 - » IP restriction / Service endpoints
- Employees / Partners
 - » Connect over internet
 - » Conditional Access

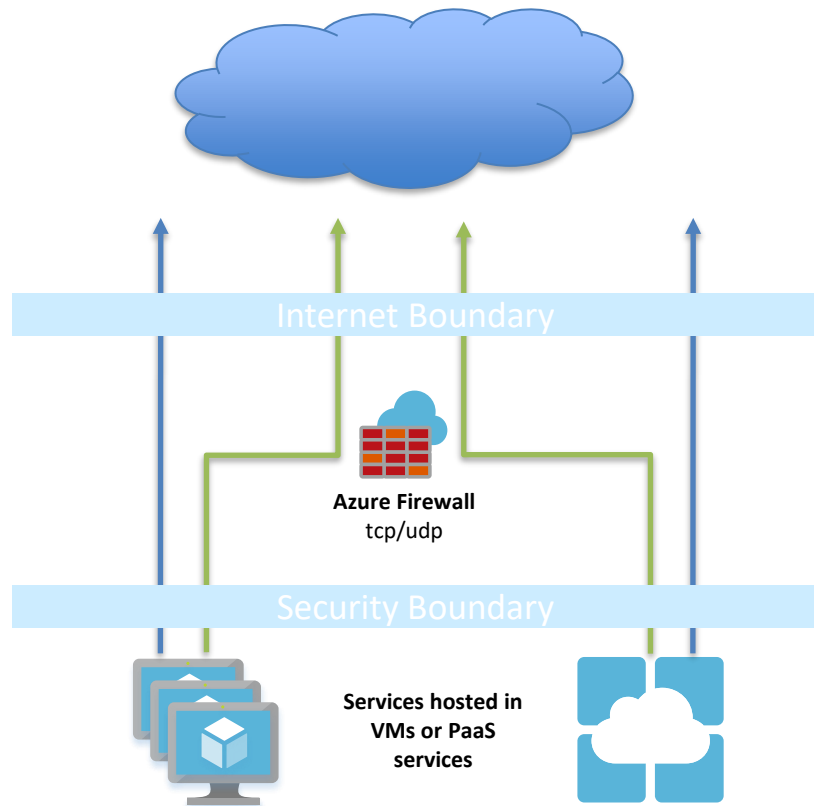


Egress traffic

- IaaS virtual network
 - » Directly
 - » Azure Firewall
- PaaS public service
 - » Directly
 - » VNET integrated through Azure Firewall

Discussion

Should Ingress, Egress and transit traffic be separated (firewall per type of traffic)?



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LandingZone options for Canopus

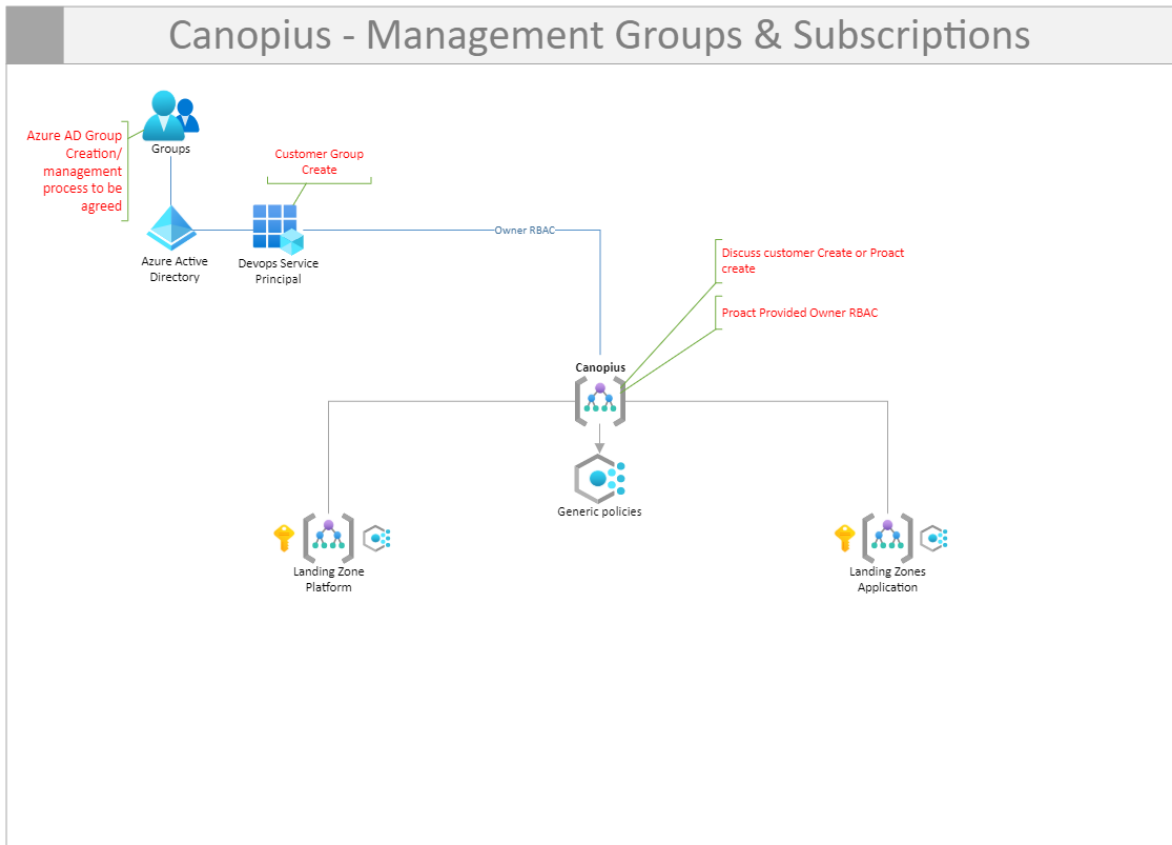


LZ Management

Discussion LandingZone

Who will manage?

Who will have rights?; to
deploy, to read

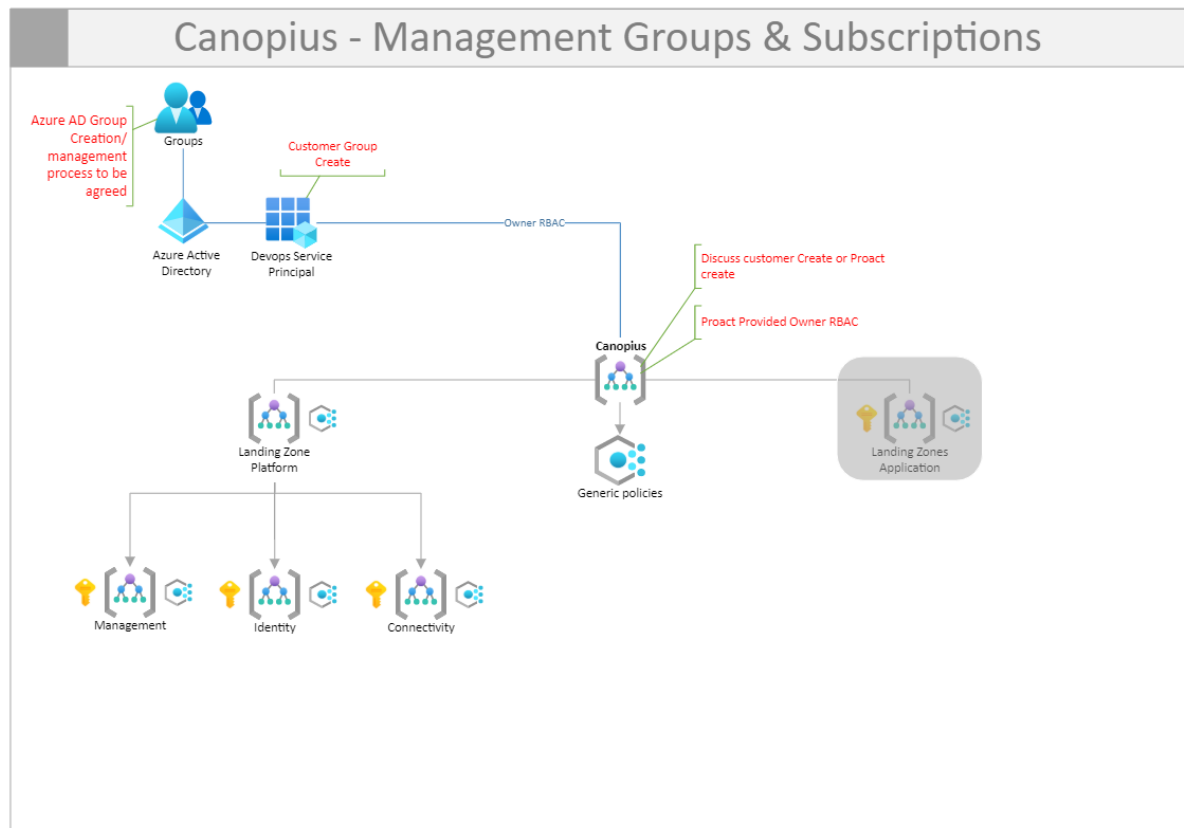


LZ Management

Discussion LZ Platform

Who will manage?

Who will have rights?; to
deploy, to read



LZ Management

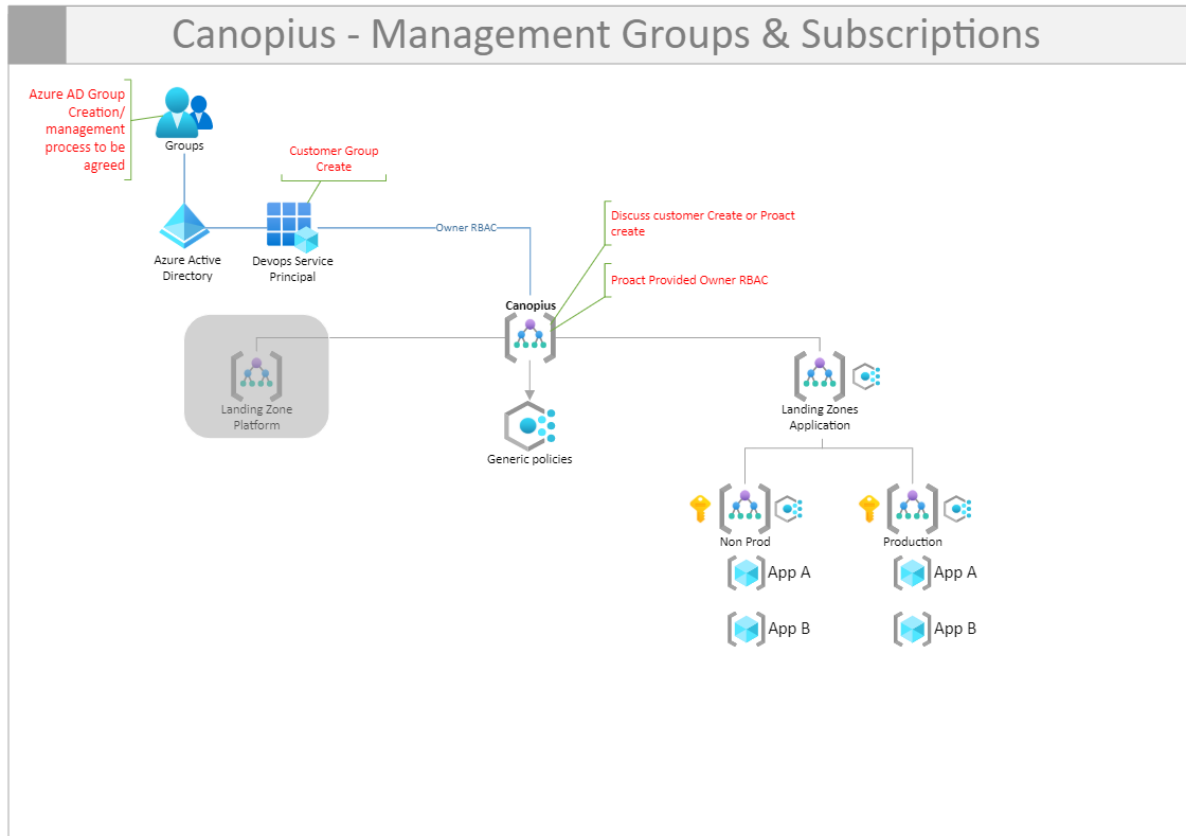
Discussion LZ Application

Devide Prod <> Non-Prod in subscriptions; enables granular control:

- RBAC
- Cost Management
- Policies

Who will manage?

Who will have rights?; to deploy, to read



LZ Management

Discussion LZ Application

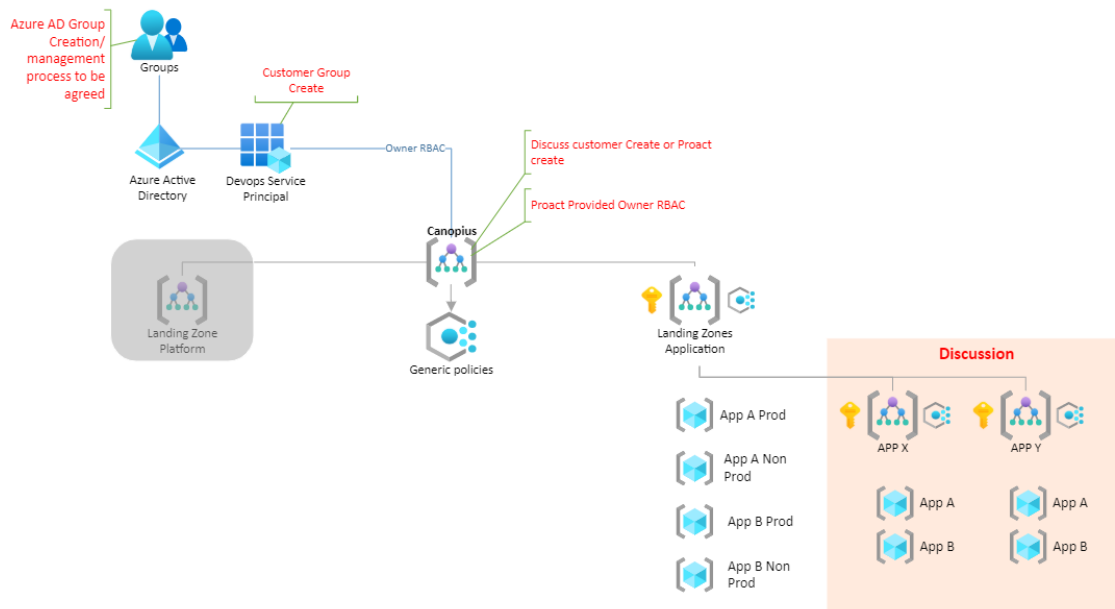
Devide Prod <> Non-Prod in subscriptions

Extra subscriptions for large applications (fi. DataWarehouse, AVD)

Who will manage?

Who will have rights?; to deploy, to read

Canopus - Management Groups & Subscriptions



LZ Network

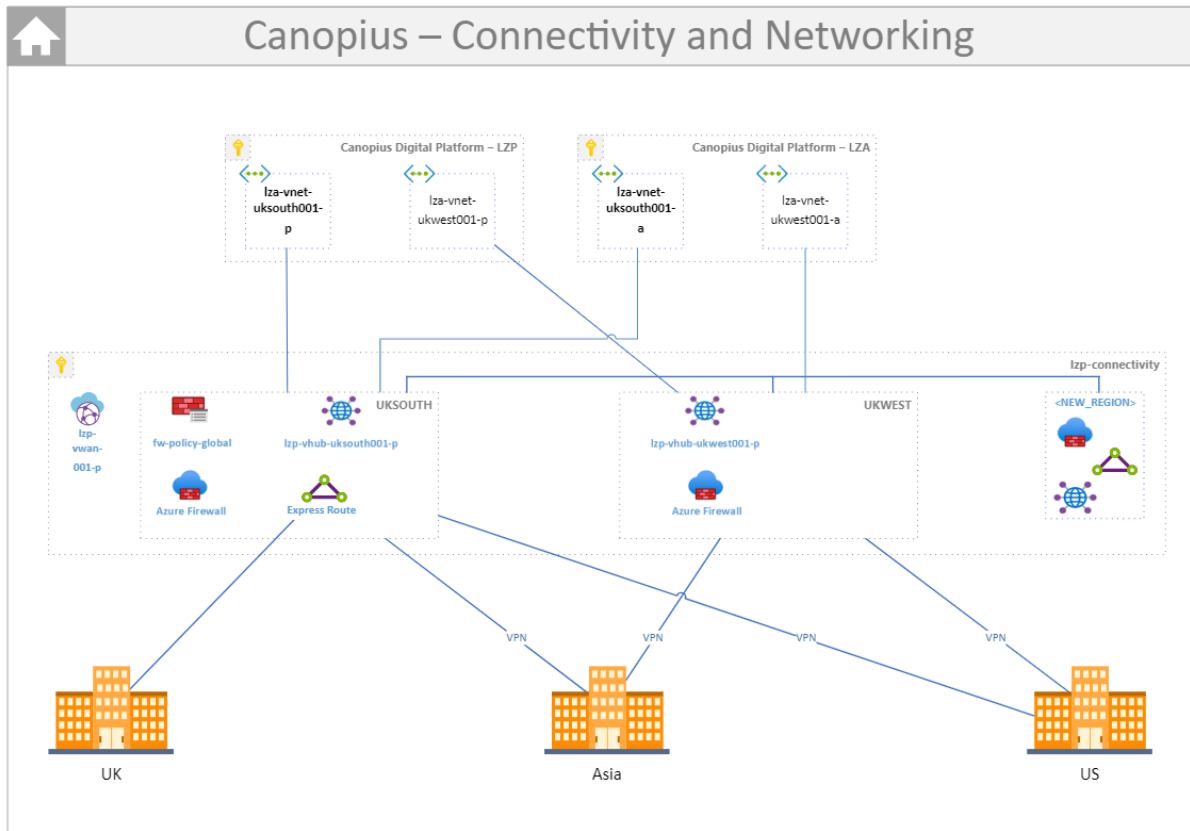
Discussion LZ Network

Use of Expressroute and
VPN

Who will manage?

Which external parties will
be involved?

Which global locations of
Canopus will participate?



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LZ Regions

Discussion LZ Application

Example:

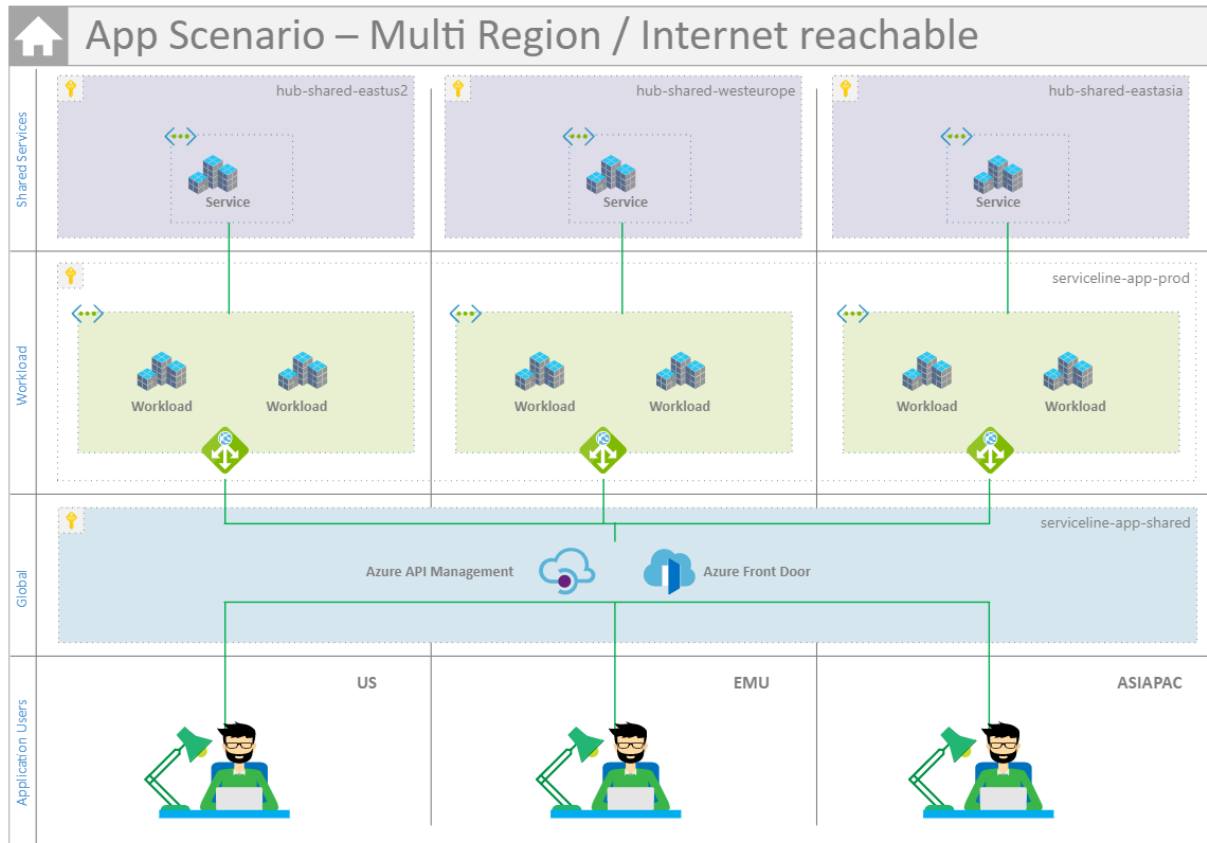
Use of multiple regions

Geo location

Azure Front Door

Region performance;

Improving user experience



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Discussion



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